

Question 4. Movie Recommendations and Cluster Analysis [set by Robert Loftin - 10 marks]

You should finish the lab session on collaborative filtering before solving part A, and the lab session on clustering before solving part B.

Data:

Use **wget** to download the [MovieLens 20M Dataset](#) to the “Data” folder and unzip it there. Please read the [dataset description](#) to understand the data before completing the following tasks.

Tasks:

A. Time-split Recommendations [4 marks]:

1. Perform time-split recommendation using ALS-based matrix factorisation in PySpark on the rating data in **ratings.csv**:

- **sort** all data by the timestamp,
- perform **splitting according to the sorted timestamp. Earlier times (the past) should be used for training, and later times (the future) should be used for testing**, which is a more realistic setting than random splits. Consider three such splits with training data sizes of 40%, 60%, and 80%.

[1 mark]

For **each** of the three splits above, study two versions (*settings*) of ALS using your student number (keeping only the digits) as the seed for the following

- **Setting 1:** The same ALS setting you used in Lab 6, except for the random seed
- **Setting 2:** Based on the results (*see the next step 3 below*) from the first ALS setting, choose a **different ALS setting that can potentially improve the results**. Provide at least a **one-sentence justification to explain** why you think the chosen setting can potentially improve the results. [*This is to imagine a real scenario. You need to think about how the performance might be improved, provide a justification, and then make changes. This implies that failing to improve the results is **acceptable**, but we expect you to provide a good justification when you make changes aiming to improve the results, and that your justification is sound.*]

[2 marks]

2. For each split and each version of ALS, compute three metrics: the Root Mean Square Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Error (MAE). Put these RMSE, MSE and MAE results for each of the three splits into one **Table** for the two ALS settings in the report. You need to report $3 \text{ metrics} \times 3 \text{ splits} \times 2 \text{ ALS settings} = 18$ numbers. Visualise these 18 numbers in ONE single figure.

[1 mark]

B. User Analysis [4 marks]:

1) After ALS, **each user** is modelled by a vector of factors. For each of the three time-splits, use *k*-means in PySpark with ***k*=25** to cluster all the users based on the user factors learned with ALS **Setting 2** above, and find the top five largest user clusters. As before, use the digits of your student number as the random seed for initialising the clusters. Report the size of (i.e. the number of users in) each of the top five clusters in one **Table**, in total $3 \text{ splits} \times 5 \text{ clusters} = 15$ numbers. Visualise these 15 numbers in ONE single figure.

[2 marks]

2) For each of the three splits in Q4-A1, consider only the *largest* user cluster in Q4-B1, and do the following on the *training* set only:

- Considering **all users** in the largest user cluster, find all the movies that have been rated by these users and their respective average ratings (over users in this cluster), and name this collection as *movies_largest_cluster*. Find those movies in *movies_largest_cluster* with an average rating greater than or equal to 4 (≥ 4), and name these as *top_movies*.
- Use **movies.csv** to find the **genres** for all of the *top_movies* and report the top ten most popular genres. *Each movie may have multiple genres, separated by the character '|'. Here “most popular” means genres assigned to the largest number of top_movies*). Report these $3 \text{ splits} \times 10 \text{ genres} = 30$ genres in one **Table**.

[2 marks]

- C. Discuss your two most interesting observations from A & B above, each with three sentences: 1) What is the observation? 2) What are the possible causes of the observation? 3) How useful is this observation to a movie website such as **Netflix**? Your report must be clearly written, and your code must be well-documented so that it is clear what each step is doing.

[2 marks]

The END of the Assignment